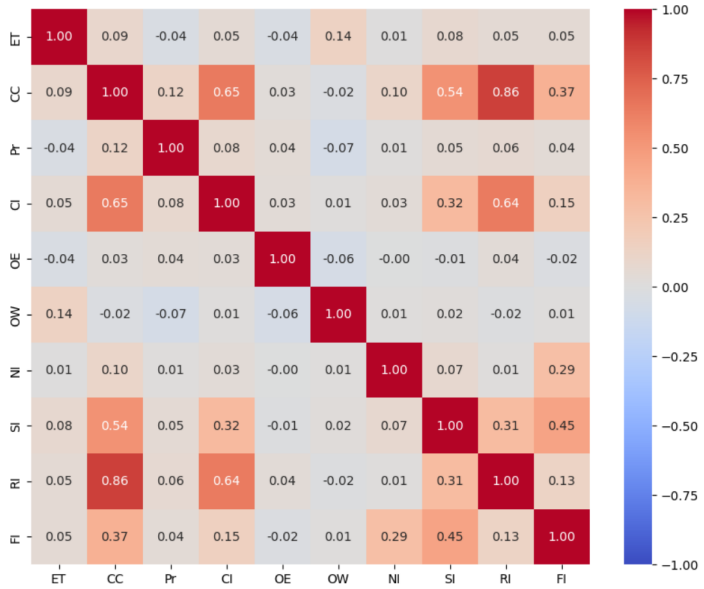


FIGURE A1 Correlation plot for the OpenStack dataset

□

APPENDIX

A PAIRWISE CORRELATIONS

Figures A1– A3 show Pearson correlations among the study variables for OpenStack, Eclipse and Android.

The correlogram in Figure A1 displays the Pearson correlation coefficients among the variables, with colour coding. Two patterns stand out. First, CC correlates strongly with several other variables — RI (0.86), CI (0.65), and FI (0.37) — reflecting that bugs attracting more contributors also tend to accumulate more repeated, community, and fresh interactions; this mutual entanglement is precisely why CC is treated as a descendant of RI (and excluded from the adjustment set) rather than as an independent covariate in our causal model (Section **??**). Second, ET shows only weak pairwise correlations with every other variable (all $|r| < 0.14$, including RI at 0.05 and SI at 0.08), despite RI and SI later emerging as statistically significant causal drivers of ET once OW and OE are adjusted for (Section refsec:estimate). This divergence between weak marginal correlation and non-trivial adjusted causal effect illustrates the core motivation for this study: raw correlation is a poor guide to causal relevance once confounders are properly conditioned on, and the correlogram should be read as evidence of interdependence among predictors rather than as a preview of the causal estimates.

The correlogram in Figure A2 displays the Pearson correlation coefficients among the variables, with colour coding. As in OpenStack, CC is strongly associated with RI (0.71) and CI (0.76), consistent with bugs that draw many contributors also accumulating many repeated exchanges. Two Eclipse-specific patterns are worth noting relative to OpenStack: OW correlates positively with ET (0.39), suggesting that developer workload has a more visible marginal association with resolution time in Eclipse than in OpenStack (where OW–ET was 0.14); and NI–SI (0.55) and NI–FI (0.90) are both markedly higher than their OpenStack counterparts, indicating that new-developer interactions in Eclipse are more tightly coupled to fresh and seasoned interaction activity. As with OpenStack, RI and SI show only modest marginal correlation with ET (0.06 and 0.11, respectively) despite both proving statistically significant in the adjusted causal estimates of Section **??**, reinforcing that the causal signal for these variables is only recoverable after controlling for OW and OE rather than being visible in the raw correlation structure.

The correlogram in Figure A3 displays the Pearson correlation coefficients between the variables, with colour-coding. Android's correlation structure differs from OpenStack and Eclipse in two respects that anticipate the null causal findings reported in Section **??**. First, OW and OE are negatively correlated (–0.62), the only negative pairing of this magnitude across the three ecosystems; this is consistent with Android's centralised, subsystem-based triage model (Section **??**), in which highly

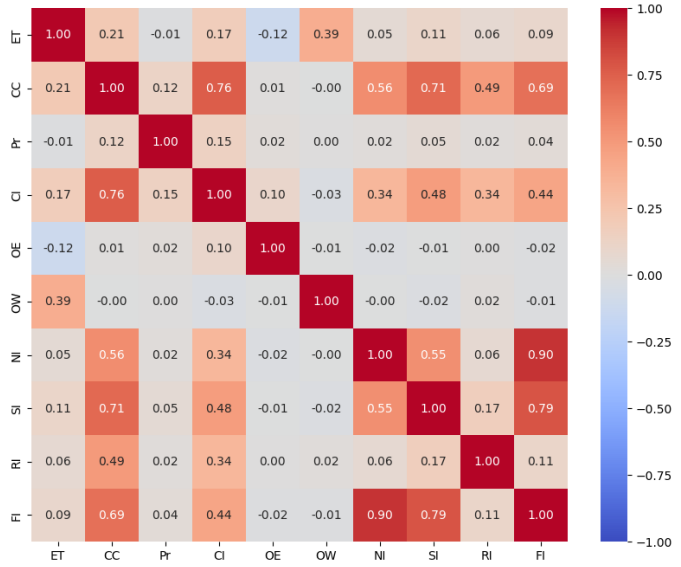


FIGURE A2 Correlation plot for the Eclipse dataset



FIGURE A3 Correlation plot for the Android dataset

engaged owners (high OE) tend to carry lighter individual bug loads (low OW), rather than the more organic, loosely coupled workload patterns seen in OpenStack and Eclipse. Second, CC is extremely tightly coupled to NI (0.94), FI (0.89), and, to a lesser extent, SI (0.36) — substantially higher than the corresponding OpenStack and Eclipse coefficients — suggesting that in Android, contributor count is almost mechanically determined by new and fresh interaction activity, leaving comparatively little independent variation for RI or SI to explain once CC-driven variation is accounted for.